Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$s = r\theta$$

Trig

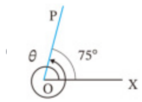
$$\sin\theta = \frac{y}{r}, \quad \cos\theta = \frac{x}{r}$$

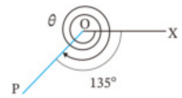
Eng Eslam Ahmed | 01120009622

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q001 Written

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus $75°$; (b) two full clockwise revolutions and a further $135°$.



(b)


Solution

1. Add $360° + 75°$ for (a).
2. Use $360° \times (-2) - 135°$ for (b).

Final answer

$(a)\ 435° \quad (b)\ -855°$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q002 Written

Draw the terminal side for $300°$ and $-450°$.

Solution

1. Reduce each angle by complete turns while keeping the sign of rotation.
2. $300°$ ends in quadrant IV; $-450°$ is coterminal with $-90°$.

Final answer

$300° : QIV \quad -450° : negative\ y - axis$

Q003 Written

Express $420°$ and $-210°$ as $\alpha + 360° \times n$, where n is an integer and $0° \leq \alpha < 360°$.

Solution

1. Choose the coterminal angle in the interval $0° \leq \alpha < 360°$.
2. Write the full family by adding $360° \times n$.

Final answer

$60° + 360° \times n \quad ; \quad 150° + 360° \times n$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of $30°$ from the positive x-axis; (b) a terminal side $40°$ above the negative x-axis, measured counter-clockwise from the positive x-axis).

2 Draw the terminal side for the following angles.

Solution

1. Clockwise rotation is negative, so (a) is $-30°$.
2. For (b), $180° - 40° = 140°$.

Final answer

$(a)\ -30° \quad (b)\ 140°$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q005 Written

Draw the terminal side for $500°$ and $-120°$.

Solution

1. Subtract or add complete turns: $500° - 360° = 140°$; $-120°$ is already in QIII.

Final answer

$500° : QI \quad -120° : QIII$

Q006 Written

Express $520°$ and $-140°$ as $\alpha + 360° \times n$.

Solution

1. Reduce each angle to the principal representative in $[0°, 360°)$.
2. Attach $360° n$.

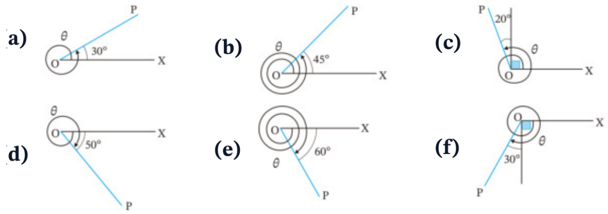
Final answer

$160° + 360° \times n \quad ; \quad 220° + 360° \times n$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q007 Written

Find the six diagram measures θ shown in the source figure (a)–(f).



2 Draw the terminal side for the following angles.

Solution

1. Read the marked acute angle and the direction of each complete turn.
2. Use $360° \pm$ marked angle and $90° \pm 20°$ where shown.

Final answer

$30°;\ 405°;\ 110°;\ 310°;\ 300°;\ 240°$

Worked Solutions   **Defining Angles and Basic Trigonometric Functions**

Q008 Written

Draw the terminal sides for $470°, 780°, 225°, -380°, -990°, -200°$.

Solution

1. Reduce each angle modulo $360°$, preserving clockwise direction for negative angles.
2. Sketch the ray from the positive x-axis in the resulting quadrant.

Final answer

QII; QII; QIII; QI; QII; QII

Q009 Written

Express $610°, 920°, 840°, -430°, -960°, -20°$ as $\alpha + 360° \times n$.

Solution

1. Find each representative α in $[0°, 360°)$.
2. The integer coefficient of $360°$ is absorbed into arbitrary integer n.

Final answer

$250° + 360°n;\ 200° + 360°n;\ 120° + 360°n;\ 290° + 360°n;\ 120° + 360°n;\ 340° + 360°n$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q010 Written

Express $129720°$ and $-2592100°$ as $\alpha + 360° \times n$, with $0° \leq \alpha < 360°$.

Solution

1. Divide by $360°$ and retain the remainder in the required interval.
2. Write each result as $\alpha + 360°n$.

Final answer

$120° + 360°n \quad ; \quad 260° + 360°n$

Q011 MCQ

The principal angle coterminal with $765°$ is:

Solution

1. $765° - 720° = 45°$

Correct option: A

Final answer

$45°$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q012 MCQ

The principal angle coterminal with -410° is:

Solution

1. $-410^\circ + 720^\circ = 310^\circ$

Correct option: C

Final answer

310°

Q013 MCQ

The terminal side of 540° lies on the:

Solution

1. $540^\circ - 360^\circ = 180^\circ$

Correct option: B

Final answer

negative x-axis

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q014 MCQ

A positive angle of 315° terminates in:

Solution

1. $315^\circ \in \mathrm{QIV}$

Correct option: D

Final answer

QIV

Q015 MCQ

Which is the general-angle form for -75°?

Solution

1. $-75^\circ + 360^\circ = 285^\circ$

Correct option: B

Final answer

$285^\circ + 360^\circ n$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q016 MCQ

The direction of a negative angle is:

Solution

1. Negative rotation is clockwise.

Correct option: B

Final answer

clockwise

Q017 MCQ

The terminal side of $1080°$ is the:

Solution

1. $1080° = 3 \times 360°$

Correct option: A

Final answer

positive x-axis

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q018 MCQ

For $\alpha = 0°$, the general-angle family is:

Solution

1. Substitute $\alpha = 0°$ in $\alpha + 360°n$.

Correct option: A

Final answer

$360°n$

Q019 MCQ

The principal angle for $-990°$ is:

Solution

1. $-990° + 1080° = 90°$

Correct option: C

Final answer

$90°$

Worked Solutions　　**Defining Angles and Basic Trigonometric Functions**

Q020 MCQ

A ray (P) is 25° above the negative x-axis. Its standard-position angle is:

Solution

1. $180^\circ - 25^\circ = 155^\circ$

Correct option: B

Final answer

155°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S.

Solution

1. Multiply degrees by $\pi/180$, and radians by $180/\pi$.
2. Use $L = r\theta$ and $S = \frac{1}{2}r^2\theta$ with θ in radians.

Final answer

$\frac{7\pi}{2};\ -315^\circ;\ L = 4\pi,\ S = 6\pi$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q022 **Written**

Complete the standard-angle degree/radian table, then: convert $288°, -400°$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Solution

1. Use the conversion multipliers and the standard table.
2. For a sector, substitute directly in $L = r\theta$ and $S = \frac{1}{2}r^2\theta$.

Final answer

$$\frac{8\pi}{5}, -\frac{20\pi}{9};\ 330°, -108°;\ L = \frac{9\pi}{2}, S = \frac{27\pi}{2}$$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q023 **Written**

Convert $108°, 36°, -240°, -225°, 420°, 400°, -570°, -2700°$ **to radians.**

Solution

1. Multiply every degree measure by $\pi/180$.
2. Reduce each fraction to lowest terms.

Final answer

$\frac{3\pi}{5}, \frac{\pi}{5}, -\frac{4\pi}{3}, -\frac{5\pi}{4}, \frac{7\pi}{3}, \frac{20\pi}{9}, -\frac{19\pi}{6}, -15\pi$

Q024 **Written**

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ **to degrees.**

Solution

1. Multiply every radian measure by $180°/\pi$.
2. Keep the sign and simplify.

Final answer

$210°, 144°, 120°, 240°, -225°, -90°, -990°, -70°$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Solution

1. Apply $L = r\theta$.
2. Apply $S = \frac{1}{2}r^2\theta$ to each sector.

Final answer

$(a)\ (7\pi, 21\pi);\ (b)\ (\frac{8\pi}{5}, \frac{16\pi}{5});\ (c)\ (\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Solution

1. Use the standard radian-angle table and the sign of rotation.
2. Sketch each terminal ray from the positive x-axis.

Final answer

QI; QI; QII; QII; QIII; QIII; QIV; QIV

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q027 Written

A sector of radius $9\,\text{cm}$ and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Solution

1. The sector arc becomes the cone base circumference: $2\pi R = 9(4\pi/3)$, so $R = 6$.
2. The slant height is 9, hence $h = \sqrt{9^2 - 6^2} = 3\sqrt{5}$.
3. Use $V = \frac{1}{3}\pi R^2 h$.

Final answer

$h = 3\sqrt{5}\,\text{cm}; \quad V = 36\pi\sqrt{5}\,\text{cm}^3$

Q028 MCQ

Convert $150°$ to radians:

Solution

1. $150 \times \pi/180 = 5\pi/6$

Correct option: A

Final answer

$$\frac{5\pi}{6}$$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:

Solution

1. $-5\pi/6 \times 180/\pi = -150°$

Correct option: B

Final answer

$-150°$

Q030 MCQ

For $r = 5, \theta = \frac{2\pi}{3}$, the arc length is:

Solution

1. $L = r\theta = 5(2\pi/3)$

Correct option: B

Final answer

$\frac{10\pi}{3}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q031 MCQ

For $r = 4, \theta = \frac{\pi}{2}$, the sector area is:

Solution

1. $S = \frac{1}{2}(4^2)(\pi/2) = 4\pi$

Correct option: B

Final answer

4π

Q032 MCQ

The radian measure of a full turn is:

Solution

1. A full circumference subtends 2π radians.

Correct option: B

Final answer

2π

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Solution

1. $7\pi/4 \times 180/\pi = 315°$

Correct option: C

Final answer

$315°$

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Solution

1. $-11\pi/6 + 2\pi = \pi/6$

Correct option: A

Final answer

$\frac{\pi}{6}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q035 MCQ

For $r = 3, \theta = \pi$, the sector area is:

Solution

1. $S = \frac{1}{2}(3^2)\pi = 9\pi/2$

Correct option: D

Final answer

$\frac{9\pi}{2}$

Q036 MCQ

The angle $\frac{5\pi}{3}$ lies in:

Solution

1. $5\pi/3 = 300°$

Correct option: D

Final answer

QIV

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q037 Written

Convert $252°$ to radians.

Solution

1. $252\pi/180 = 7\pi/5$.

Final answer

$$\frac{7\pi}{5}$$

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S.

Solution

1. $L = 8(3\pi/4) = 6\pi$.
2. $S = \frac{1}{2}(8^2)(3\pi/4) = 24\pi$.

Final answer

$$L = 6\pi, \quad S = 24\pi$$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin\theta, \cos\theta, \tan\theta$.

Solution

1. The terminal side is in QIII with reference angle $\pi/3$.
2. Use the signs of x, y on the unit circle.

Final answer

$-\frac{\sqrt{3}}{2},\ -\frac{1}{2},\ \sqrt{3}$

Q040 Written

Find the exact values (write undefined where appropriate): $\sin\frac{7\pi}{4}, \cos\frac{\pi}{3}, \tan\frac{\pi}{6}, \sin(-\frac{2\pi}{3}), \cos(-\pi), \tan(-\frac{\pi}{2})$.

Solution

1. Locate each angle on the unit circle.
2. Read x, y, then use $\sin\theta = y, \cos\theta = x, \tan\theta = y/x$.

Final answer

$-\frac{\sqrt{2}}{2};\ \frac{1}{2};\ \frac{1}{\sqrt{3}};\ -\frac{\sqrt{3}}{2};\ -1;\ undefined$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q041 **Written**

Find the exact values of the eighteen trigonometric functions listed in the source exercise (a)–(r), including negative and undefined cases.

Solution

1. Reduce each angle to a standard reference angle.
2. Apply quadrant signs and the unit-circle exact values; tangent is undefined when $x = 0$.

Final answer

$\frac{\sqrt{3}}{2}; -\frac{1}{2}; 0; -\frac{\sqrt{2}}{2}; \frac{1}{2}; -\frac{\sqrt{3}}{2}; -\sqrt{3}; 1; 0; -\frac{1}{2}; \frac{\sqrt{2}}{2}; \frac{\sqrt{3}}{2}; \frac{\sqrt{2}}{2}; -\frac{\sqrt{3}}{2}; -\frac{1}{2}; -\sqrt{3}; \frac{1}{\sqrt{3}}; undefined$

Q042 **Written**

The panel height is $h(t) = 3\sin(\frac{2\pi}{3}t - \frac{\pi}{4}) + 1.5$. Find the exact height at $t = 0$.

Solution

1. Substitute $t = 0$: $h(0) = 3\sin(-\pi/4) + 1.5$.
2. Use $\sin(-\pi/4) = -\sqrt{2}/2$.

Final answer

$\frac{3(1-\sqrt{2})}{2}$ m

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q043 MCQ

$\sin\frac{\pi}{6}$ **equals:**

Solution

1. Use the QI special-angle value.

Correct option: B

Final answer

$\frac{1}{2}$

Q044 MCQ

$\cos\frac{2\pi}{3}$ **equals:**

Solution

1. QII has negative cosine; reference angle $\pi/3$.

Correct option: B

Final answer

$-\frac{1}{2}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q045 MCQ

$\tan\frac{3\pi}{4}$ **equals:**

Solution

1. QII reference angle $\pi/4$.

Correct option: A

Final answer

-1

Q046 MCQ

$\sin(-\frac{\pi}{2})$ **equals:**

Solution

1. The terminal point is $(0, -1)$.

Correct option: A

Final answer

-1

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q047 **MCQ**

$\tan\frac{\pi}{2}$ **is:**

Solution

1. $\tan = y/x$ and $x = 0$.

Correct option: D

Final answer

undefined

Q048 **MCQ**

$\cos\frac{5\pi}{3}$ **equals:**

Solution

1. $5\pi/3$ has reference angle $\pi/3$ in QIV.

Correct option: D

Final answer

$\frac{1}{2}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q049 MCQ

If $P(x, y)$ lies on the unit circle, then $\sin\theta$ is:

Solution

1. By definition, $\sin\theta = y$.

Correct option: B

Final answer

y

Q050 MCQ

$\tan(-\frac{3\pi}{4})$ **equals:**

Solution

1. Tangent is odd: $\tan(-3\pi/4) = -\tan(3\pi/4) = 1$.

Correct option: C

Final answer

1

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q051 MCQ

$\sin \frac{11\pi}{6}$ **equals:**

Solution

1. QIV reference angle $\pi/6$, so sine is negative.

Correct option: A

Final answer

$-\frac{1}{2}$

Q052 Written

Find $\sin \frac{5\pi}{6}, \cos \frac{4\pi}{3}, \tan \frac{7\pi}{4}$.

Solution

1. Use reference angles $\pi/6, \pi/3, \pi/4$ and quadrant signs.

Final answer

$\frac{1}{2}, -\frac{1}{2}, -1$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q053 **Written**

Find $\sin(-\frac{5\pi}{6}), \cos(-\frac{7\pi}{6}), \tan(-\frac{\pi}{3})$.

Solution

1. Reduce negative angles and apply odd/even symmetry.

Final answer

$-\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}$

Q054 **Written**

For $P = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ **on the unit circle, find the corresponding** $\sin\theta, \cos\theta, \tan\theta$.

Solution

1. Read x, y directly, then compute y/x.

Final answer

$\frac{\sqrt{3}}{2}, -\frac{1}{2}, -\sqrt{3}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q055 **Written**

Find the exact value of $3\sin\frac{3\pi}{2} - 2\cos\pi$.

Solution

1. $3(-1) - 2(-1) = -1$.

Final answer

-1

Q056 **Written**

State the two standard angles where $\tan\theta$ is undefined in $[0, 2\pi)$.

Solution

1. Tangent is undefined when the unit-circle x-coordinate is zero.

Final answer

$\frac{\pi}{2}, \frac{3\pi}{2}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q057 **Written**

Solve for θ, $0 \le \theta < 2\pi$: (a) $\sin\theta = -\frac{1}{2}$; (b) $\cos\theta = \frac{1}{\sqrt{2}}$; (c) $\tan\theta = -\sqrt{3}$.

Solution

1. Draw the unit circle and the corresponding horizontal/vertical/oblique line.
2. Use the reference angle and select the quadrants with the required sign.

Final answer

$(a)\ \frac{7\pi}{6}, \frac{11\pi}{6};\ (b)\ \frac{\pi}{4}, \frac{7\pi}{4};\ (c)\ \frac{2\pi}{3}, \frac{5\pi}{3}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q058 **Written**

Solve the six trigonometric equations in the source Try set for $0 \le \theta < 2\pi$: $\sin\theta = 1/\sqrt{2}$, $\cos\theta = -\sqrt{3}/2$, $\tan\theta = 1/\sqrt{3}$, $\sin\theta = 0$, $\cos\theta = 1/2$, $\tan\theta = -1$.

Solution

1. Find the reference angle from the special triangle.
2. Use the sign of the function to choose the correct quadrants.

Final answer

$\frac{\pi}{4}, \frac{3\pi}{4}; \frac{5\pi}{6}, \frac{7\pi}{6}; \frac{\pi}{6}, \frac{7\pi}{6}; 0, \pi; \frac{\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{4}, \frac{7\pi}{4}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q059 Written

Solve the sixteen equations in the source Exercise set (a)–(p) for $0 \le \theta < 2\pi$.

Solution

1. Apply the same unit-circle/reference-angle method to each equation.
2. Remember that sine/cosine can have two solutions, while $\sin\theta = \pm 1$ and $\cos\theta = \pm 1$ have one in the interval.

Final answer

$\frac{\pi}{6}, \frac{5\pi}{6}; \frac{\pi}{3}, \frac{2\pi}{3}; \frac{\pi}{2}, \frac{3\pi}{2}; 0;\ 0, \pi;\ \frac{\pi}{3}, \frac{4\pi}{3}; \frac{\pi}{2}; \frac{5\pi}{4}, \frac{7\pi}{4}; \frac{3\pi}{4}, \frac{5\pi}{4}; \frac{2\pi}{3}, \frac{4\pi}{3}; \frac{5\pi}{6}, \frac{11\pi}{6}; \frac{\pi}{4}, \frac{5\pi}{4}; \frac{4\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{2}; \pi; \frac{\pi}{6}, \frac{11\pi}{6}$

Q060 Written

Solve $\tan^2\theta + (1-\sqrt{3})\tan\theta = \sqrt{3}$ for $0 \le \theta < 360°$.

Solution

1. Let $u = \tan\theta$ and factor $u^2 + (1-\sqrt{3})u - \sqrt{3} = 0$.
2. The two tangent values are 1 and $-\sqrt{3}$; list their angles in $[0, 360°)$.

Final answer

$60°, 135°, 240°, 315°$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q061 MCQ

For $0 \le \theta < 2\pi$, $\sin\theta = \frac{1}{2}$ gives:

Solution

1. Sine is positive in QI and QII with reference angle $\pi/6$.

Correct option: A

Final answer

$\frac{\pi}{6}, \frac{5\pi}{6}$

Q062 MCQ

For $0 \le \theta < 2\pi$, $\cos\theta = -\frac{1}{2}$ gives:

Solution

1. Cosine is negative in QII and QIII.

Correct option: B

Final answer

$\frac{2\pi}{3}, \frac{4\pi}{3}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q063 MCQ

For $0 \le \theta < 2\pi$, $\tan\theta = 1$ gives:

Solution

1. Tangent is positive in QI and QIII.

Correct option: A

Final answer

$\frac{\pi}{4}, \frac{5\pi}{4}$

Q064 MCQ

The solutions of $\sin\theta = 0$ in $[0, 2\pi)$ are:

Solution

1. The unit-circle y-coordinate is zero on the x-axis.

Correct option: A

Final answer

$0, \pi$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q065 MCQ

If $\cos\theta = 1$, then in $[0, 2\pi)$, θ is:

Solution

1. The point $(1, 0)$ corresponds to $\theta = 0$.

Correct option: A

Final answer

0

Q066 MCQ

For $0 \leq \theta < 2\pi$, $\tan\theta = -\sqrt{3}$ gives:

Solution

1. Tangent is negative in QII/QIV; reference angle $\pi/3$.

Correct option: B

Final answer

$\frac{2\pi}{3}, \frac{5\pi}{3}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin\theta = -1$ gives:

Solution

1. The lowest unit-circle point has y-coordinate -1.

Correct option: B

Final answer

$\frac{3\pi}{2}$

Q068 MCQ

For $0 \leq \theta < 2\pi$, $\cos\theta = 0$ gives:

Solution

1. Cosine is the x-coordinate, zero on the y-axis.

Correct option: B

Final answer

$\frac{\pi}{2}, \frac{3\pi}{2}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q069 MCQ

The equation $\tan\theta = 0$ has solutions in $[0, 2\pi)$:

Solution

1. Tangent is zero when $y = 0$ and $x \neq 0$.

Correct option: A

Final answer

$0, \pi$

Q070 Written

Solve $\cos\theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Cosine is positive in QI/QIV with reference angle $\pi/6$.

Final answer

$\frac{\pi}{6}, \frac{11\pi}{6}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q071 **Written**

Solve $\sin\theta = -\frac{\sqrt{3}}{2}$ **for** $0 \le \theta < 2\pi$**.**

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Final answer
$\frac{4\pi}{3}, \frac{5\pi}{3}$

Q072 **Written**

Solve $\tan\theta = \frac{1}{\sqrt{3}}$ **for** $0 \le \theta < 2\pi$**.**

Solution

1. Tangent is positive in QI/QIII with reference angle $\pi/6$.

Final answer
$\frac{\pi}{6}, \frac{7\pi}{6}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Q073 **Written**

Solve $2\cos\theta - 1 = 0$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Rearrange to $\cos\theta = 1/2$, then choose QI/QIV.

Final answer

$\frac{\pi}{3}, \frac{5\pi}{3}$

Q074 **Written**

Solve $2\sin\theta + \sqrt{2} = 0$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Rearrange to $\sin\theta = -\sqrt{2}/2$, then choose QIII/QIV.

Final answer

$\frac{5\pi}{4}, \frac{7\pi}{4}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Quiz MCQ 1 Concept Quiz · MCQ

The radian measure of $240°$ is:

Solution

1. $240\pi/180 = 4\pi/3$.

Correct option: B

Final answer

$\frac{4\pi}{3}$

Quiz MCQ 2 Concept Quiz · MCQ

If $\sin\theta = -\frac{\sqrt{3}}{2}$, $0 \le \theta < 2\pi$, then θ is:

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Correct option: B

Final answer

$\frac{4\pi}{3}, \frac{5\pi}{3}$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Quiz MCQ 3 **Concept Quiz · MCQ**

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Solution

1. $S = \frac{1}{2}(6^2)(\pi/3) = 6\pi$.

Correct option: C

Final answer

6π

Quiz MCQ 4 **Concept Quiz · MCQ**

The general-angle family for $-300°$ is:

Solution

1. Add $360°$ to obtain the principal angle $60°$.

Correct option: A

Final answer

$60° + 360° n$

Worked Solutions **Defining Angles and Basic Trigonometric Functions**

Quiz WRITTEN 5 Concept Quiz · Written

Solve $\cos\theta = -\frac{\sqrt{2}}{2}$ for $0 \le \theta < 2\pi$.

Solution

1. Cosine is negative in QII/QIII with reference angle $\pi/4$.

Final answer
$\frac{3\pi}{4}, \frac{5\pi}{4}$

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Solution

1. Multiply by $180/\pi$: $-390°$. Add $720°$ for the principal representative $330°$.

Final answer
$-390°;\ 330°$

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Solution

1. $L = 10(2\pi/5) = 4\pi$.
2. $S = \frac{1}{2}(10^2)(2\pi/5) = 20\pi$.

Final answer

$L = 4\pi, \quad S = 20\pi$

Quiz WRITTEN 8 Concept Quiz · Written

Evaluate $3\sin\frac{7\pi}{6} - 2\cos\frac{4\pi}{3}$.

Solution

1. $3(-1/2) - 2(-1/2) = -1/2$.

Final answer

$-\frac{1}{2}$

Answer key

Use the question number shown on each page.

Q001 $(a)\ 435°\quad (b)\ -855°$

Q002 $300°\ :\ QIV$

$-450°\ :\ negative\ y-axis$

Q003 $60° + 360° \times n\quad ;\quad 150° + 360° \times n$

Q004 $(a)\ -30°\quad (b)\ 140°$

Q005 $500°\ :\ QI\quad -120°\ :\ QIII$

Q006 $160° + 360° \times n\quad ;\quad 220° + 360° \times n$

Q007 30°; 405°; 110°; 310°; 300°; 240°

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250° + 360°n;\ 200° + 360°n;\ 120° + 360°n;\ 290° + 360°n;\ 120° + 360°n;\ 340° + 360°n$

Q010 $120° + 360°n\quad ;\quad 260° + 360°n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2};\ -315°;\ L = 4\pi,\ S = 6\pi$

Q022 $\frac{8\pi}{5}, -\frac{20\pi}{9};\ 330°, -108°;\ L = \frac{9\pi}{2}, S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}, \frac{\pi}{5}, -\frac{4\pi}{3}, -\frac{5\pi}{4}, \frac{7\pi}{3}, \frac{20\pi}{9}, -\frac{19\pi}{6}, -15\pi$

Q024 $210°, 144°, 120°, 240°, -225°, -90°, -990°, -70$

Q025 $(a)\ (7\pi, 21\pi);\ (b)\ (\frac{8\pi}{5}, \frac{16\pi}{5});\ (c)\ (\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}\,\text{cm};\quad V = 36\pi\sqrt{5}\,\text{cm}^3$

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi,\quad S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2},\ -\frac{1}{2},\ \sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2};\ \frac{1}{2};\ \frac{1}{\sqrt{3}};\ -\frac{\sqrt{3}}{2};\ -1;\ undefined$

Q041 $\frac{\sqrt{3}}{2}; -\frac{1}{2}; 0; -\frac{\sqrt{2}}{2}; \frac{1}{2}; -\frac{\sqrt{3}}{2}; -\sqrt{3}; 1; 0; -\frac{1}{2}; \frac{\sqrt{2}}{2}; \frac{\sqrt{3}}{2}; \frac{\sqrt{2}}{2}; -\frac{\sqrt{3}}{2}; -\frac{1}{2}; -\sqrt{3}; \frac{1}{\sqrt{3}}; undefined$

Q042 $\frac{3(1-\sqrt{2})}{2}\ \text{m}$

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}, -\frac{1}{2}, -1$

Q053 $-\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}, -\frac{1}{2}, -\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}, \frac{3\pi}{2}$

Q057 $(a)\ \frac{7\pi}{6}, \frac{11\pi}{6};\ (b)\ \frac{\pi}{4}, \frac{7\pi}{4};\ (c)\ \frac{2\pi}{3}, \frac{5\pi}{3}$

Q058 $\frac{\pi}{4}, \frac{3\pi}{4};\ \frac{5\pi}{6}, \frac{7\pi}{6};\ \frac{\pi}{6}, \frac{7\pi}{6};\ 0, \pi;\ \frac{\pi}{3}, \frac{5\pi}{3};\ \frac{3\pi}{4}, \frac{7\pi}{4}$

Q059 $\frac{\pi}{6}, \frac{5\pi}{6};\ \frac{\pi}{3}, \frac{2\pi}{3};\ \frac{\pi}{2}, \frac{3\pi}{2};\ 0;\ 0, \pi;\ \frac{\pi}{3}, \frac{4\pi}{3};\ \frac{\pi}{2};\ \frac{5\pi}{4}, \frac{7\pi}{4};\ \frac{3\pi}{4}, \frac{5\pi}{4};\ \frac{2\pi}{3}, \frac{4\pi}{3};\ \frac{5\pi}{6}, \frac{11\pi}{6};\ \frac{\pi}{4}, \frac{5\pi}{4};\ \frac{4\pi}{3}, \frac{5\pi}{3};\ \frac{3\pi}{2};\ \pi;\ \frac{\pi}{6}, \frac{11\pi}{6}$

Q060 $60°, 135°, 240°, 315°$

Q061 A

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}, \frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}, \frac{5\pi}{3}$

Q072 $\frac{\pi}{6}, \frac{7\pi}{6}$

Q073 $\frac{\pi}{3}, \frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}, \frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}, \frac{5\pi}{4}$

Quiz WRITTEN 6 $-390°;\ 330°$

Quiz WRITTEN 7 $L = 4\pi,\quad S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$